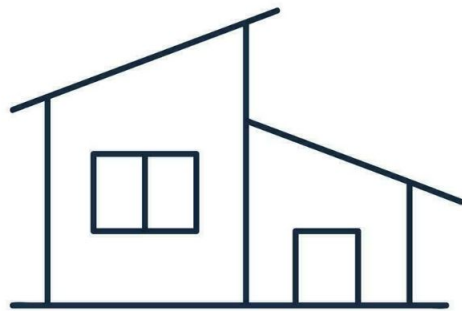


Forecasting
Housing
Outcomes with
\$250k



The \$250k Question: Buy or Rent?

\$250,000



- You have \$250,000 to invest in your pocket
- Two choices: Buy a home or rent & invest
- Which of this path will be more profitable?
- Varies by city, time horizon, market

The Challenge: Making the Right Financial Decision



Interest Rates
Fluctuate, impacting affordability



Property Values
Vary by location, market trends

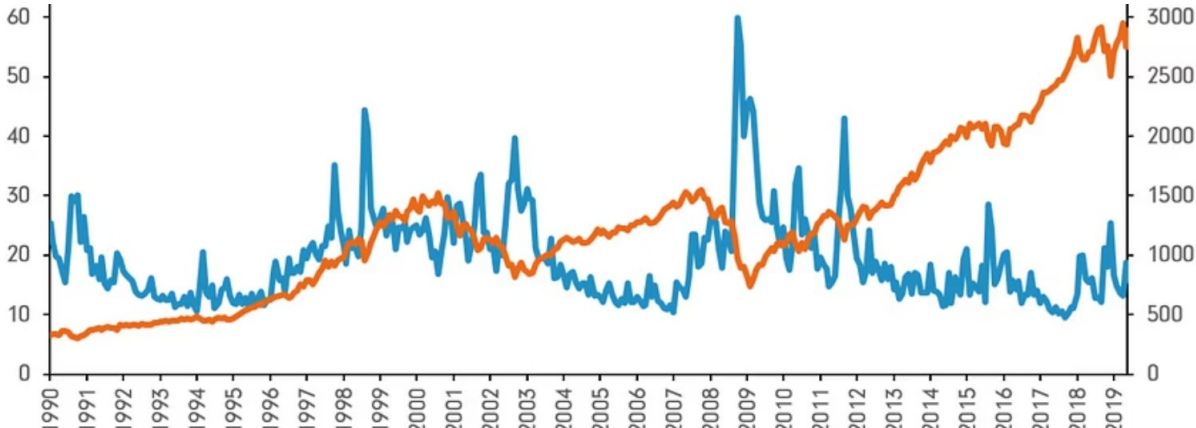


Rent Costs
Dynamic, influence budget

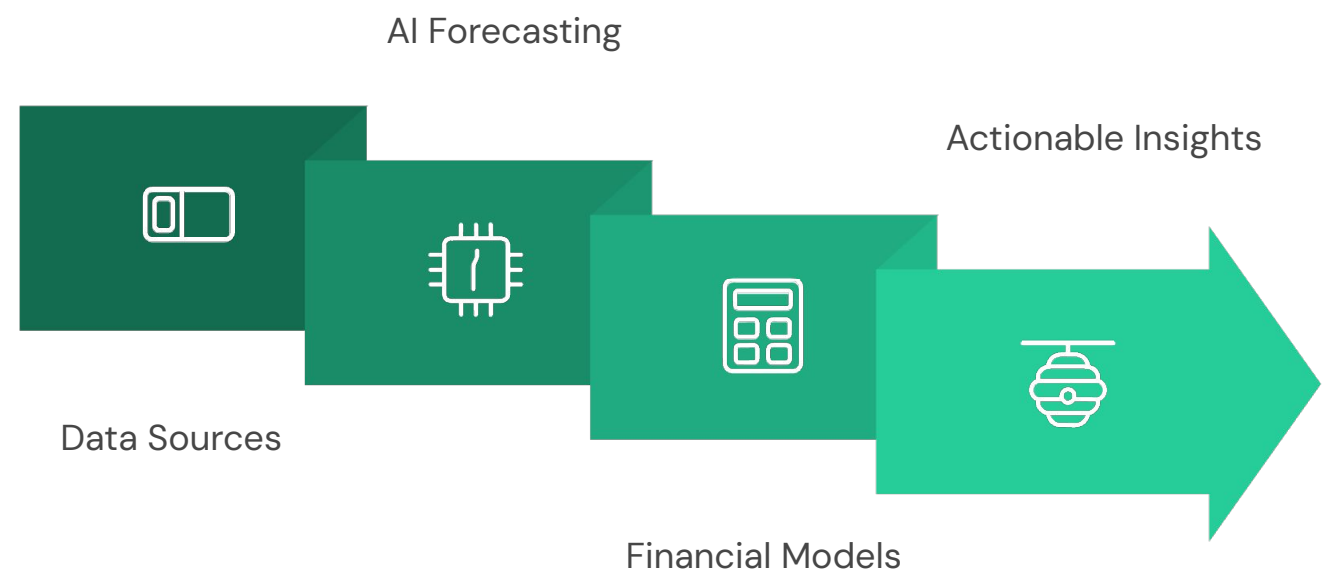


Investment Returns
Opportunity cost of capital

Traditional advice is often one-size-fits-all, but market conditions change rapidly. Investors need data-driven insights, not gut feelings.



Our Data-Driven Approach



1

Data Sources
Zillow, FRED API for mortgage rates

2

AI Forecasting
ARIMA models for prices & rates

3

Financial Modeling
NPV, IRR, break-even analysis

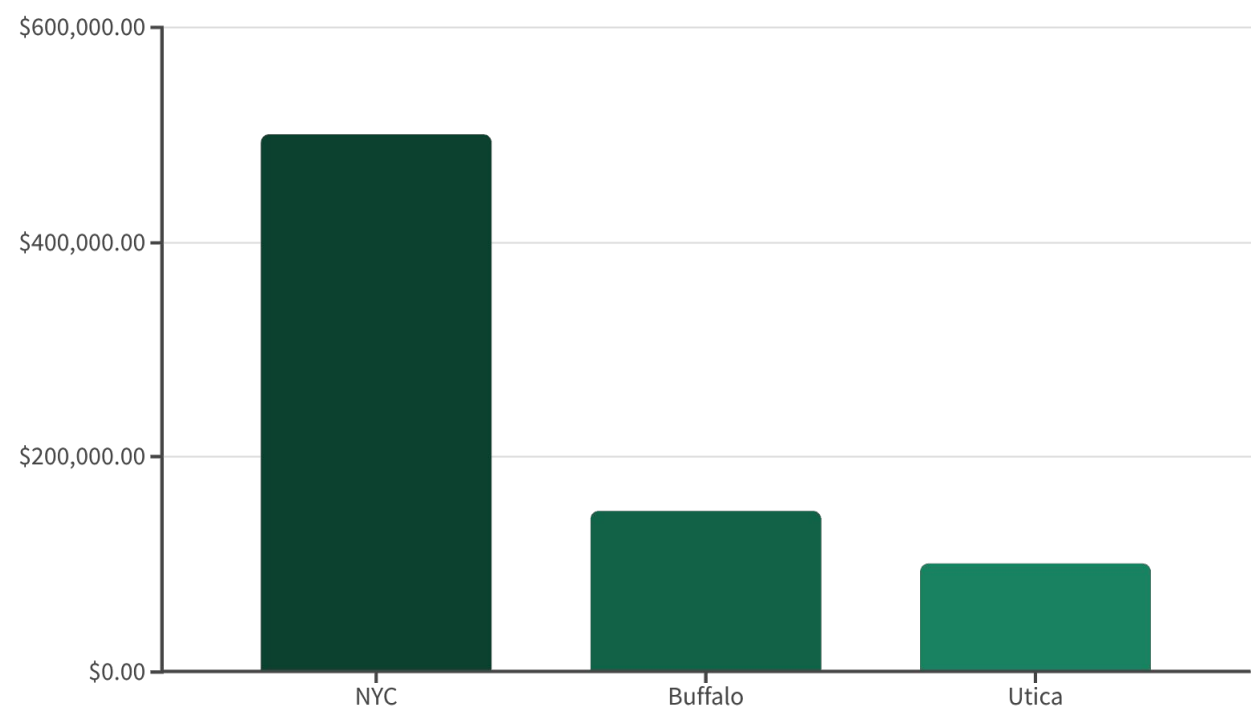
4

Multi-City Analysis
NYC, Buffalo, and Utica explored

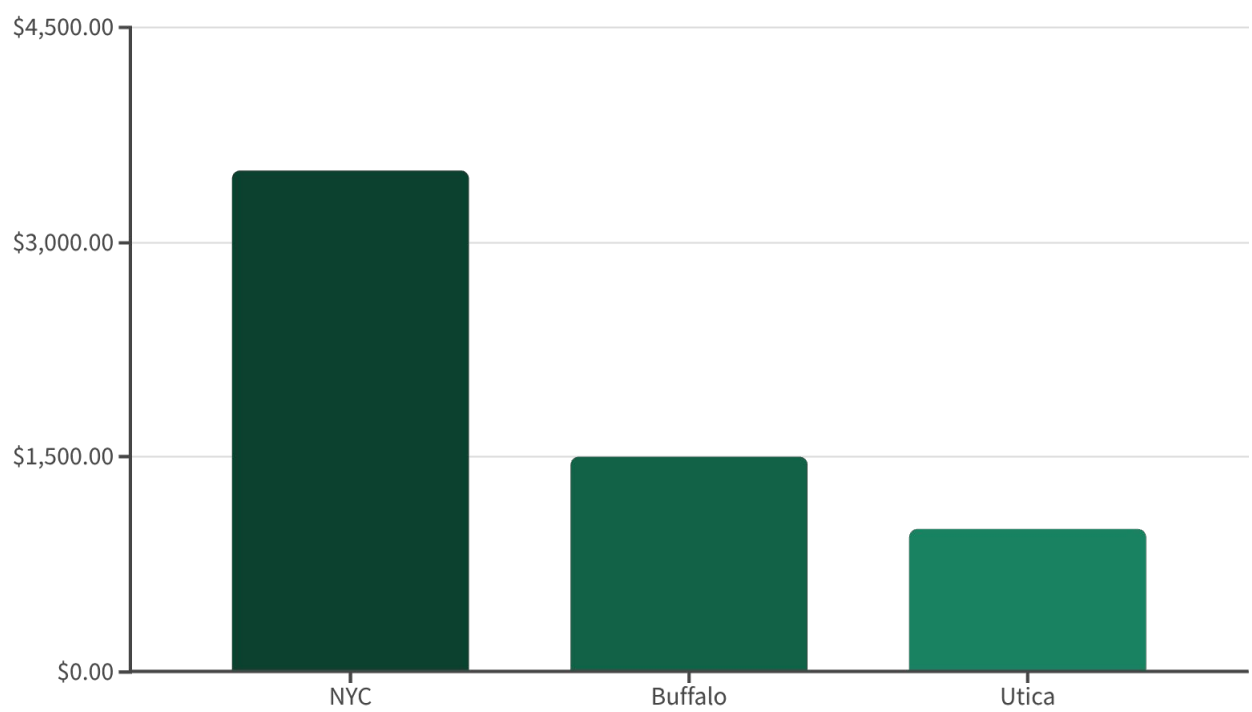
Current Market Conditions: What the Data Shows

Mortgage Rates: Around 6–7% (30-year fixed)

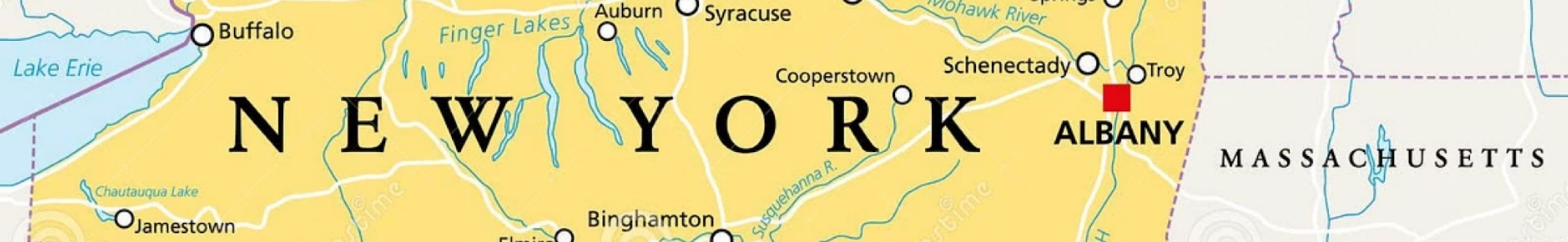
Property Values: Vary Dramatically



Rental Costs: Follow Similar Patterns



Market Volatility: Rates and prices change monthly, impacting buy vs. rent decisions.



The Answer Depends on Where You Live

1

NYC

Rent (\$69K savings, +\$180K wealth)

2

Buffalo

Rent (\$56K savings, +\$364K wealth)

3

Utica

Rent (\$50K savings, +\$350K wealth)

10-Year Horizon: Renting Wins in All Cities

→ 30-Year Horizon:

- NYC: Buying wins after 19.25 years
- Utica: Buying wins after 13.75 years
- Buffalo: Buying wins after 21.3 years

→ Key Takeaway:

Time horizon impacts decision more than city

Buying v Renting in America

Source: Zillow / Case Shiller / BLS



When Does Buying Make Financial Sense?

Break-Even Point: Where cumulative buying costs = cumulative renting costs

1	NYC Month 231 (19.25 years) January 2045
2	Utica Month 165 (13.75 years) July 2039
3	Buffalo Month 256 (21.3 years) February 2047

Before Break-Even:

Renting is cheaper

After Break-Even:

Buying becomes cheaper & advantage grows

30-Year Wealth Projection: The Long-Term View

Why Buying Wins Long-Term

- Mortgage paid off
- House appreciation (243% forecast)
- Rent continues to rise (122% increase)

Why Renting Wins Short-Term

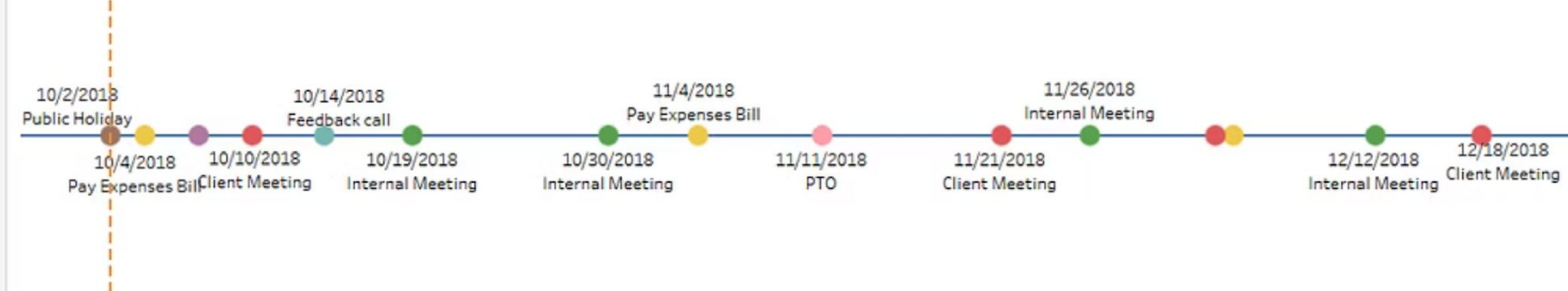
- Investment returns compound faster
- Lower upfront costs
- More flexibility

30-Year Final Wealth Comparison

NYC: Buying advantage of ~\$24K

10-Year Final Wealth Comparison

- NYC: Renting advantage of ~\$180K
- Utica: Renting advantage of ~\$350K
- Buffalo: Renting advantage of ~\$364K



Interactive Dashboard: Explore the Data Yourself



Real-Time Exploration

Filter by city, time, strategy



Visual Insights

Break-even, wealth, costs



Customizable Views

Compare cities side-by-side



Data-Driven Decisions

Insights backed by models

Accessible to non-technical users for informed decision-making.

Our Predictions vs. Reality: Model Validation

- ARIMA Model Selection
Chosen for realistic mortgage rate forecasts
- Historical Accuracy
Models trained on years of historical data
- Confidence Intervals
Forecasts include uncertainty ranges
- Continuous Improvement
Models update with new data

Our models capture market trends and provide actionable insights, validated against historical patterns.

- Income Calculation

